Astronomy Test

CSE 2025

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Instructions: Circle one letter for each multiple choice question. Answer free response questions in the space provided below each question. Questions are worth one point unless noted otherwise.

1. Which of the following correctly lists the standard Harvard spectral classification scheme in order of increasing temperature?

A) OBAFGKM
B) MKGFABO
C) ABFGKMO
D) OMKGFBA
E) OMGFABK

2. Fill in the blank: The Chandrasekhar limit is the _____

A) Maximum mass of a stable neutron star
B) Minimum mass of a black hole
C) Minimum mass of a star that will go supernova
D) Maximum mass of a stellar black hole
E) Maximum mass of a stable white dwarf star

3. In which region of the Hertzsprung-Russell diagram are we most likely to find stars in hydrostatic equilibrium?

4. List the quantum mechanical principle/theory responsible for the electron degeneracy pressure that keeps white dwarfs from collapsing under their own gravity.

5. Which of these is closest to the average mass of (known) white dwarfs in the Milky Way?

A) Less than 0.4 Solar Masses
B) 0.4 to 0.9 Solar Masses
C) 0.9 to 2 Solar Masses
D) 2 to 10 Solar Masses
E) More than 10 Solar Masses

6. Which of the following stellar properties is the MOST important determinant for the length of a star's lifetime?

A) Mass
B) Temperature
C) Radius
D) Metallicity

7. Which of the following elements are considered “metals” in astronomy?

I. Lithium II. Oxygen III. Silicon IV. Iron

A) I, II, III, IV
B) I, III, IV
C) II, III, IV,
D) III, IV
E) Only IV

8. Which of the following units is most ideal for parallax calculations?

A) meters
B) km
C) AU
D) light years
E) parsecs

9. Fill in the blanks: Direct imaging is best for detecting planets that have a _____ size and are _____ to/from their star.

A) small, close
B) small, far
C) large, close
D) large, far

10. The transit method is best for detecting what kind of planets?

A) Super-Earths
B) Neptunian Planets
C) Hot Jupiters
D) Gas Giants
E) Rogue Planets

11. What is the name of the process by which planetesimals gain mass? (One word)

12. How long do we have until the sun enters its red giant phase and burns Earth to a crisp? (write your answer in years)

13. Fill in the blank: As a star's temperature rises, the peak wavelength of its radiation _____.

A) increases
B) stays the same
C) decreases

14. Which of the following stars produces the highest amount of red light? (Assume constant radius)

A) An orange star
B) A yellow star
C) A blue star

15. Select the list correctly sorted from least dense to most dense

A) White Dwarf, Brown Dwarf, Neutron Star, Red Giant
B) Red Giant, White Dwarf, Brown Dwarf, Neutron Star
C) Brown Dwarf, Red Giant, White Dwarf, Neutron Star
D) Red Giant, Brown Dwarf, White Dwarf, Neutron Star
E) Brown Dwarf, Red Giant, Neutron Star, White Dwarf

16. Name the 3 main types of Galaxies (1 pt each)

17. Which of the following is an elliptical collection of tens of thousands to millions of low metallicity stars that is found in orbit around a galaxy?

A) Globular Cluster
B) Local Group
C) Open Cluster
D) Elliptical Galaxy
E) Satellite Galaxy

18. What medium do gravitational waves propagate through?

A) The Higgs Field
B) Electromagnetic Fields
C) Spacetime
D) Matter
E) The Interstellar Medium

19. What constant describes the rate of the universe's expansion?

A) Stefan-Boltzmann Constant
B) Speed of Light
C) Planck's Constant
D) Hubble Constant
E) Gravitational Constant

20. Which wavelength regime would you use to trace cool clouds of neutral hydrogen?

A) Ultraviolet
B) X-ray
C) Radio
D) Infrared
E) Visible

21. There is a cutoff mass at which stars will end their lives in core-collapse supernova, and this mass is closest to:

A) $4M_\odot$
B) $9M_\odot$
C) $15M_\odot$
D) $30\ M_\odot$
E) There is no cutoff mass, silly!

22. Fill in the blank: Earth's average temperature is _____ what is predicted by planetary equilibrium temperature calculations.

A) hotter than
B) colder than
C) almost exactly

23. Which of the following is NOT a standard method for obtaining the distance of a celestial object?

A) Cepheid Variables
B) Type 1a Supernovae
C) Redshift
D) Radial Velocity
E) Parallax

24. Which of the following are YOU not in?

A) Solar System
B) Andromeda Galaxy
C) The universe
D) Virgo Supercluster
E) Local group

25. Fill in the blank: Neutron stars are kept from collapse by _____.

A) Thermonuclear Fusion
B) Pauli Exclusion Principle
C) Magnetic Fields
D) Centrifugal forces
E) Neutron Degeneracy Pressure

26. Fill in the blank: The Oort Cloud encompasses the outer regions of the _____.

A) Atmosphere
B) Solar System
C) Milky Way
D) Virgo Supercluster
E) Observable Universe

27. Which of the following is NOT a hypothetical candidate for dark matter?

A) MACHOs
B) Axions
C) WIMPs
D) Supermassive Black Holes
E) Sterile Neutrinos

28) Which of the following is NOT a type of quark? ;)

A) Up
B) Bottom
C) Left
D) Charm
E) Strange

29. Fill in the blank: The Sun is NOT _____

A) A star
B) 15 million ^{o}C at its core
C) A G-Type main sequence star
D) An orange dwarf
E) 1.99 * 10^30 kilograms

30. What is the characteristic feature of an H II region in terms of its spectral lines?

A) Continuum emission without lines
B) Strong absorption lines
C) Emission of neutral hydrogen
D) Emission lines from ionized hydrogen

31. What phenomenon is typically associated with Herbig-Haro objects?

A) Spiral arm formation in galaxies
B) Supernova remnants
C) Outflows from young stars
D) Accretion into a black hole

32. What is the main source of heat energy in a protostar?

A) Electromotive Forces
B) Gravitational Potential Energy
C) Deuterium Fusion
D) Proton-Proton Chain

Math Section:

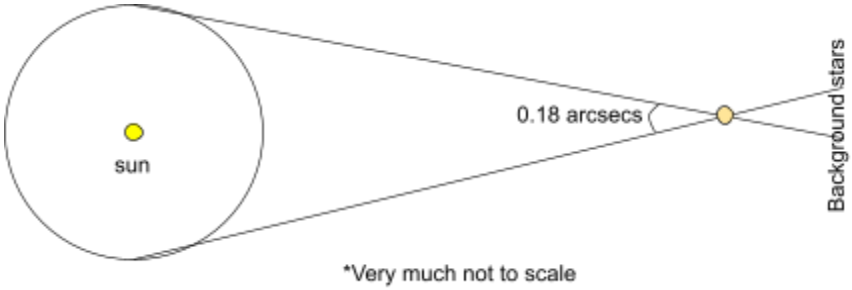
Questions are worth 4 points each. Show sufficient work.

33. A Strömgren radius is the radius around a star in which hydrogen is ionized by the star's radiation. The radius can be described by the following formula:

$$R_s = (\frac{3Q_s}{4\pi n^2 \alpha})^x$$

R is the length of the Strömgren radius, *Q* is the number of energetic photons emitted in a unit of time, n is the number density of hydrogen atoms, and α is the recombination coefficient with units of meters^3 per second.
x is an unknown real number constant. What must x be so that the units match up?
(3 points)

34. A team of astronomers observe a K type star and find that over the course of 6 months it shifts by 0.18 arcseconds compared to the background stars. Calculate the star's distance in parsecs to two decimal places. Show your steps.



35. On average, Earth receives approximately 1360 Watts per square meter of energy from the sun. Neptune has an orbital radius 29.1 AU larger than Earth's. Calculate the intensity of the Sun's light on Neptune in W/m^2. Make sure to show your work.

36. Earth has collapsed into a black hole! Derive an expression for the radius of its event horizon in terms of M (the mass of the Earth) and physical constants as appropriate. Hint: start by using this equation for escape velocity!

